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Inter-layer Information Similarity Assessment of Deep Neural Networks Via Topological Similarity and Persistence Analysis of Data Neighbour Dynamics

Andrew Hryniowski[1,2,3] **and Alexander Wong**[1,2,3]
[1] Vision and Image Processing Research Group, University of Waterloo
[2] Waterloo Artificial Intelligence Institute
[3] DarwinAI Corp.
{apphryni, a28wong}@uwaterloo.ca

Abstract

The quantitative analysis of information structure through a deep neural network (DNN) can unveil new insights into the theoretical performance of DNN architectures. Two very promising avenues of research towards quantitative information structure analysis are: 1) layer similarity (LS) strategies focused on the inter-layer feature similarity, and 2) intrinsic dimensionality (ID) strategies focused on layer-wise data dimensionality using pairwise information. Inspired by both LS and ID strategies for quantitative information structure analysis, we introduce two novel complimentary methods for inter-layer information similarity assessment premised on the interesting idea of studying a data sample's neighbourhood dynamics as it traverses through a DNN. More specifically, we introduce the concept of **Nearest Neighbour Topological Similarity** (NNTS) for quantifying the information topology similarity between layers of a DNN. Furthermore, we introduce the concept of **Nearest Neighbour Topological Persistence** (NNTP) for quantifying the inter-layer persistence of data neighbourhood relationships throughout a DNN. The proposed strategies facilitate the efficient inter-layer information similarity assessment by leveraging only local topological information, and we demonstrate their efficacy in this study by performing analysis on a deep convolutional neural network architecture on image data to study the insights that can be gained with respect to the theoretical performance of a DNN.

1 Introduction

Deep neural networks (DNNs) are functions that map information from one domain to another [1]. These maps often consist of hundreds of sub-maps in the form of element-wise non-linear functions, matrix multiplications, convolutions, etc. [1]. Each one of these sub-maps gradually warps the underlying manifold of a dataset. Studying the properties of these sub-maps and the effects on a dataset's manifold through a DNN at a micro and macro level can lead to a better understanding of a DNN's internal workings. Therefore, such a quantitative analysis of information structure through a DNN can unveil new insights into the theoretical performance of DNN architectures.

Two very promising avenues of research towards quantitative information structure analysis are: 1) intrinsic dimensionality (ID) [2, 3] strategies, and 2) layer similarity (LS) strategies [4, 5, 6, 7]. At the micro level, intrinsic dimensionality (ID) strategies allow for approximations of a manifold's dimensionality. Lacking from ID analysis is a notation of distance between layers. Knowing the number of dimensions required to represent a manifold does not illuminate the manifold's internal characteristics, and directly comparing the magnitude of the ID between layers provides limited

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actionable information. On the macro level, layer similarity (LS) strategies [4, 5, 6, 7] are designed to compare the similarity of information representations between layers. LS measures work by comparing the features of one layer to all the other features of another layer across a set of input data. As such, measuring how a local region of the dataset manifold changes is not possible.

Inspired by both LS and ID strategies for quantitative information structure analysis, we propose a data-centric approach to study the effects a DNN has on the local topological structure of a dataset's manifold based on data neighbourhood dynamics as it traverses through a DNN. More specifically, we construct nearest neighbour graph (NNG) representations to capture the topological structure of a dataset's information representation for each layer in a DNN. Then we compare each layers' NNG using two novel forms of analysis: 1) **Nearest Neighbour Topological Similarity** (NNTS) to measure the local topological similarity between layers, and 2) **Nearest Neighbour Topological Persistence** (NNTP) to investigate inter-layer interacts on a pairwise data sample basis. These two proposed approaches open the door for fine-grained analysis of the complex data dynamics present within a DNN. At a high level these methods compare the first degree relations between dataset samples within a layer to such relations in another layer.

2 Nearest Neighbour Topological Similarity

Below is a brief definition of the nearest neighbour graph (NNG) used within this work to capture properties of a dataset's topological structure. See Appendix A for the motivation behind the graph's design choices. Let $\mathbf{x}_i \in \mathbf{X}$ be a set of input samples, and let G represent a DNN. Let the output of some sub-function v_v for the $\mathbf{x}_i$ sample be defined as $\mathbf{y}_{vi} = v_v(\mathbf{x}_i; G_v)$, where $G_v \subseteq G$ contains all required sub-functions, edges, and weights to calculate $\mathbf{y}_{vi}$. The main idea behind our approach is to use a graph of neighbours to capture the local structure between samples within a layer. For a given layer v_v with a set of outputs $\mathbf{Y}_v$, let $H_v = (\mathbf{Y}_v, D_v)$ be the graph of neighbours for layer v_v, where $\mathbf{Y}_v = v_v(\mathbf{X}; G_v)$ are the vertices of the graph, and D_v are the edges between two given samples $\mathbf{y}_{vi}, \mathbf{y}_{vj} \in \mathbf{Y}_v$. Let $K_{vi} \subseteq \mathbf{Y}_v$ be an ordered set of nearest neighbours of sample $\mathbf{y}_{vi}$. Directed edges are used for NNG construction.

Let $Q(H_a, H_b) = q_{ab}$ measure the Nearest Neighbour Topological Similarity (NNTS) between layers v_a and v_b where $\mathbf{y}_{ai}$ and $\mathbf{y}_{bi}$ are sample $\mathbf{x}_i$'s representation in layers v_a and v_b, respectively. To compare a single sample across layers we propose a sample-wise similarity function $Q_s(\mathbf{y}_{ai}, \mathbf{y}_{bi})$ where $Q(\cdot)$ is a function of all $Q_s(\cdot)$. Let $Q(H_a, H_b)$ be defined as

$$Q(H_a, H_b) = \frac{1}{n} \sum_{i}^{n} Q_s(\mathbf{y}_{ai}, \mathbf{y}_{bi}) \tag{1}$$

Then for a given sample $\mathbf{x}_i$ for layers a and b we get neighbour sets K_{ai} and K_{bi}, respectively, for some given k. Let the per-sample inter-layer similarity function be defined as the IOU between layers. Note that this formulation also allows a sample to have different neighbours between layers.

$$Q_s(\mathbf{y}_{ai}, \mathbf{y}_{bi}) = \frac{|K_{ai} \cap K_{bi}|}{|K_{ai} \cup K_{bi}|} \tag{2}$$

$Q(\cdot)$ uses local information through first degree relations of a sample within a layer, and compares samples between layers by comparing the local characteristics of different representations of a sample.

We apply the notion of Nearest Neighbour Topological Similarity (NNTS) to a LeNet-5 [8] architecture trained on the MNIST [9] dataset to see how the local topological structure of a dataset changes across the model. Since LeNet-5 is a small architecture we break up what is normally considered a layer into their respective atomic operations before applying NNTS. We measure the NNTS between all pairs of operations in the LeNet-5 model. The results for NNTS analysis are shown in Figure 1. We show the NNTS matrix for four different values of k, 15, 100, 6000, and 12000. The table headers along the top and left indicate the operation with the LeNet-5 model. The operations are causally aligned moving from left to right along the top, and top to bottom on the left. I stands for the input layer, C for convolutional operations, R for ReLU activation, P for max-pooling, M for matrix multiplication, and O for output (note that O is also a matrix multiplication operation). The number following the operation identifier indicates the layer number.

Each of the NNTS matrices are symmetric about the diagonal. The diagonal in each plot all have values of 1 since layers are all self similar. Notice the block-like pattern staggered both vertically

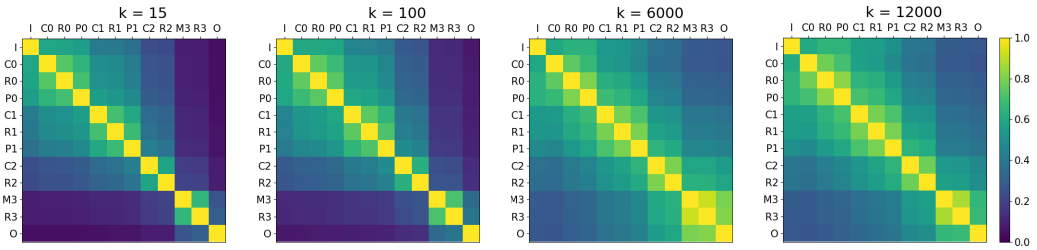

Figure 1: The Nearest Neighbour Topological Similarity between layers in a LeNet-5 model trained on the MNIST dataset for a different number of k nearest neighbours.

and horizontally in all four plots every time the layer number changes (e.g., moving from P0 to C1, or from R3 to O). This is a clear indication that sequences of operations which are normally considered *layers* (within LeNet-5) are not arbitrary since internal representations within a layer are more similar to one another than to operations outside the layer. The inter-layer similarity pattern even persists when comparing the first operation in a layer to the first operation in the following layer. This observation could be used to study other standard layer designs (e.g., a layer designed with batchnorm) to determine if such designs follow the same inter-layer similarity pattern. Notice that the similarity between P1 and M3 is marginally smaller (about 0.05) when compared the similarity between R2 and M3. The marginal difference is a good indication that removing layer 2 will have little effect on network performance.

As the number of k nearest neighbours increases from $k = 15$ to $k = 6000$ there is a gradual increase in the similarity between all layer pairs. The transition between from $k = 6000$ to $k = 12000$ sees a decrease in similarity, and most noticeably between the last couple of operations (bottom right hand corner), in the LeNet-5 model. This decrease is to be expected considering that MNIST has ten classes with approximately 6000 samples per class. Near the end of the network samples from the same class should be clustered near one another. At a $k = 6000$ a sample's connections will mostly consist of all samples from within the class. Any operation performed on samples from the same class would likely have the same effect and thus not effect the inter-layer neighbour relations. But when $k = 12000$ half a sample's neighbours will be from other classes. While operations are unlikely to effect intra-class neighbours, they can still effect inter-class neighbours, and thus resulting in the decrease in similarity from $k = 6000$ to $k = 12000$. It is expected that the inter-layer similarity converges to 1 as the number of connections approaches the number of samples in the dataset.

3 Nearest Neighbour Topological Persistence

Reducing the similarity between two layers to a single value provides a useful measure for high level measure for topological similarity. On the other hand, such reduction also removes most of the local inter-sample relationship information, thereby reducing one's ability to study the complex interactions between layers throughout a network. In this section we introduce an approach from which higher order analysis can be performed. Specially, we investigate when pairs of samples become neighbours in a DNN, properties of the pairs while they are neighbours, and when pairs of samples are no longer neighbours. We call such analysis Nearest Neighbour Topological Persistence (NNTP).

Consider a network where layers follow a sequential design $v_{in} > \cdots > v_a > \cdots > v_b > \cdots > v_{out} \in V$, where layer v_{in} is the input layer and v_{out} is the output layer of a DNN. Let e_{ij} be an abstract un-directed connection between samples $(\mathbf{x}_i, \mathbf{x}_j)$, and let e^v_{ij} be the un-directed connection between samples $(\mathbf{x}_i, \mathbf{x}_j)$ in layer v. Let $e^v_{ij} \in H_v$ iff either of the corresponding directional connections are in H_v, where H_v is the NNG of layer v.

Let e_{ij} be α-$persistent$ between two layers v_a and v_b if there exists no more than α contiguous layers in the chain of layers $v_a > \cdots > v_b$ where $e^v_{ij} \notin H_v$ for all v's within the chain of layers. α-$persistent$ is a whole family of measures. In this work we only investigate 0-$persistent$ sample pairs. 0-$persistent$ can be interpreted as a measure for local network stability. If a connection persists across a series of layers then it is reasonable to assume that the pair is located in a local region of the dataset's manifold that share specific features. Analysing when samples are no longer neighbours may help illuminate what specific features a given network layer is detecting. When considering the